EE1007
Introduction to Computer Science Laboratory

Lab 3
Control Statement I

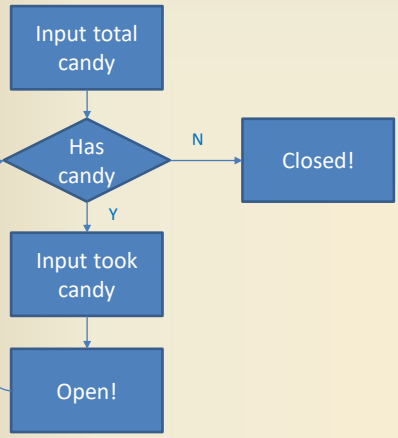
Andy, Yu-Guang Chen
Assistant Professor, Department of EE
National Central University
andyygchen@ee.ncu.edu.tw

Lab 3.1
P113009 Candy Bar (30%)

- Test target: while loop
- There is a candy bar with a limited amount of candy. Once all the candy has been taken, the bar will close.
- You are asked to write a program that informs the user whether the candy bar is still open or closed. If the candy bar still has candy, the program should print, "The candy bar is open!" The program will keep running until there is no more candy, at which point it should print, "The candy bar is closed!"

Lab 3.1
P113009 Candy Bar (30%)

◆思考方向



Lab 3.1
P113009 Candy Bar (30%)

◆ Control opening time

```
while (total > 0) {
    cout << "The candy bar is open!" << endl;
    cout << "How many candy do you want to take: ";
    cin >> candy;
    total = total - candy;

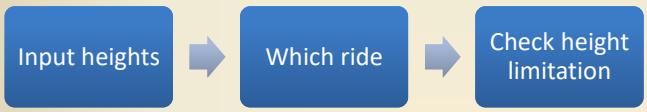
}
```

Lab 3.2
P113010 Height Limitation (30%)

- Goal: if…else
- You are asked to write a program that determines how many of the three people meet the height requirements and can take the ride.
- First, the program will ask for the heights of the 3 people. Then, the user will select the ride they want to take. Finally, the program will display how many of the 3 people meet the height requirements and can take the ride.

Lab 3.2
P113010 Height Limitation (30%)

- 思考方向



Lab 3.2
P113010 Height Limitation (30%)

- Decide how many people can take the ride

```
if (r == 1) {
    if (h1 < 140)
        p -= 1;

    if (h2 < 140)
        p -= 1;

    if (h3 < 140)
        p -= 1;
}

if (r == 2) {
    if (h1 < 120)
        p -= 1;

    if (h2 < 120)
        p -= 1;

    if (h3 < 120)
        p -= 1;
}

if (r == 3) {
    if (h1 < 100)
        p -= 1;

    if (h2 < 100)
        p -= 1;

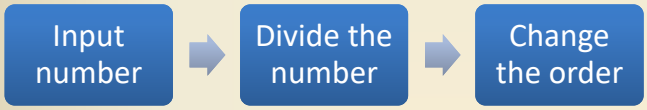
    if (h3 < 100)
        p -= 1;
}
```


Lab 3.3
P113011 Palindrome (30%)

- Test target: if…else, while loop
- A palindromic number is a number that remains the same when its digits are reversed, such as 16461.
- You are asked to write a program to check whether the input number is palindrome or not.

Lab 3.2
P113010 Height Limitation (30%)

◆思考方向



Lab 3.3
P113011 Palindrome (30%)

◆ Decide palindrome

```
    temp = num;

    while (temp != 0) {
        digit = temp % 10;
        rev = rev * 10 + digit;
        temp = temp / 10;
    }

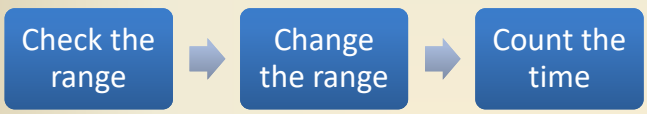
    if (rev != num)
        cout << "NO" << endl;
    else
        cout << "YES" << endl;
```

Lab 3.4
P113012 Find the Bomb (20%)

- Goal: if…else, while loop
- You are asked to write a program to count how many rounds the soldier used to find the bomb.
- The bomb is hidden between 0 to 100, but not including 0 and 100. First, the screen will print “Enter the bomb number (0~100): ”. After entering the target, the screen will print “Range 0~100: ” and then the soldier starts to guess the number. If entering 65 and the bomb is at 50, the screen will print “Range 0~65: “. When the soldier finds the bomb, the screen will show the bomb number and the rounds that the soldier found the bomb. If the number entering out of range, shows "Out of range! Please enter again: ".

Lab 3.4
P113012 Find the Bomb (20%)

- 思考方向



Lab 3.4
P113012 Find the Bomb (20%)

- Target/guess range

```
while (guess <= down || guess >= up) {
    cout << "Out of range! Please enter again: ";
    cin >> guess;
}
```

- Change range

```
if (guess < target)
    down = guess;
else if (guess > target)
    up = guess;
```


Evaluation

- Lab3.1(30%)
  - 5 testcases, 6% each
- Lab3.2(30%)
  - 5 testcases, 6% each
- Lab3.3(30%)
  - 5 testcases, 6% each
- Lab3.4(20%)
  - 5 testcases, 4% each





Andy, Yu-Guang Chen

Assistant Professor, Department of EE, NCU

Email: andyygchen@ee.ncu.edu.tw